Solved Exercise 12.2

1. Find the arithmetic mean of each data set.

(i) For 4,6,10,12,15,20,25,28,30: $\Sigma x=150$ and $n=9$, so $\bar{x}=\frac{150}{9}=16.666\ldots$.

16.67

(ii) For 12,18,19,0,−19,−18,−12: $\Sigma x=12+18+19+0-19-18-12=0$ and $n=7$, so $\bar{x}=\frac{0}{7}=0$.

(iii) For 6.5,11,12.3,9,8.1,16,18,20.5,25: $\Sigma x=126.4$ and $n=9$, so $\bar{x}=\frac{126.4}{9}=14.0444\ldots$.

14.04

(iv) For 8,10,12,14,16,20,22: $\Sigma x=102$ and $n=7$, so $\bar{x}=\frac{102}{7}=14.5714\ldots$.

14.57, each to two decimal places where approximate.

2. Find the median height of 55,53,54,58,60,61,62,56,57,52,51,63 inches.

Sorted: 51,52,53,54,55,56,57,58,60,61,62,63. Median = $\frac{56+57}{2}$

= 56.5 inches.

3. For earnings 88,70,72,125,115,95,81,90,95,90 rupees, find mean, median and mode.

$\Sigma x=921, n=10 \Rightarrow$ mean=92.1.

Sorted: 70,72,81,88,90,90,95,95,115,125

$\Rightarrow$ median=$\frac{90+90}{2}=90$.

Both 90 and 95 occur twice, more often than every other value.

Mean=Rs. 92.1; median=Rs. 90; modes=Rs. 90 and Rs. 95 (bimodal).

4. For marks 15–19,20–24,25–29,30–34,35–39 with frequencies 9,18,35,17,5, find mean and median.

Midpoints x=17,22,27,32,37; $\Sigma f=84$ and $\Sigma fx=153+396+945+544+185=2223$.

Direct mean=$\frac{2223}{84}=26.4643$. Short check with A=27: $\Sigma fD=-90-90+0+85+50=-45$, so $\bar{x}=27-\frac{45}{84}=26.4643$.

Median = $24.5+[\frac{42-27}{35}]\times 5$

= 26.6429.

Mean≈26.46 marks; median≈26.64 marks.

5. Find the mode for classes 5–9,10–14,15–19,20–24,25–29 with frequencies 1,8,18,11,2.

Modal class = 15–19: L

= 14.5,h

= 5,f

= 18,f_p

= 8,f_n

= 11.

Mode = $14.5+[\frac{18-8}{36-8-11}]\times 5$

= $14.5+\frac{50}{17}$

Mode≈17.44.

6. Find mean, median and mode of wages 4250,4350,4400,4250,4350,4410,4500,4300,4500,4390.

$\Sigma x=43{,}700, n=10 \Rightarrow$ mean=4,370. Short check with A=4,400: $\Sigma D=-700$, so $4{,}400-\frac{700}{10}=4{,}370$.

Sorted middle values are 4350 and 4390 $\Rightarrow$ median=$\frac{4350+4390}{2}=4{,}370$.

4250,4350 and 4500 each occur twice, so the data are trimodal.

Mean=median=Rs. 4,370; modes=Rs. 4,250, Rs. 4,350 and Rs. 4,500.

7. The mean of 45 numbers is 80. Find their sum.

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$\Sigma x = n\,\bar{x}$
$= 45 \times 80$
$= 3{,}600.$

8. For 1,4,0,7,9 find mean, median and mode.

Mean=$\frac{21}{5}$=4.2; sorted 0,1,4,7,9 gives median=4; no value repeats, so no mode.

9. For 148,145,160,157,156,160, show mode>median>mean.

Mean=$\frac{926}{6}$=154.333…; sorted 145,148,156,157,160,160 gives median=$\frac{156+157}{2}$=156.5; mode=160.

160>156.5>154.33, as required.

10. For attendance 21,15,16,18,14,17,15,12,13,11 find median, mode and mean by deviation.

Sorted: 11,12,13,14,15,15,16,17,18,21 ⇒ median=15 and mode=15.
Using D=x−20, ΣD=152−10(20)=−48. Hence $\bar{x}$=20+ΣD/n=20−48/10=15.2.

Median=15, mode=15, mean=15.2.

11. Pocket-money classes 5–10,10–15,15–20,20–25,25–30 have frequencies 12,9,18,7,4. Find median, mode and coded mean.

$n = 50$. Median class
$= 15–20: L$
$= 15,h$
$= 5,cf$
$= 21,f$
$= 18$
median $= 15+[\frac{25-21}{18}]5$
$= 16.1111.$
Mode $= 15+[\frac{18-9}{36-9-7}]5$
$= 17.25.$

Midpoints=7.5,12.5,17.5,22.5,27.5. With A=17.5,h=5, coded U=$\frac{x-A}{5}$=−2,−1,0,1,2.

$\Sigma fU = -24-9+0+7+8$
$= -18$
$\bar{x} = 17.5+5(\frac{-18}{50})$
$= 15.7.$

Median≈16.11, mode=17.25, mean=15.70 rupees.

12. Mean age of 20 boys is 13 years 4 months 5 days. Find their age sum and the mean after removing one 15-year-old.

Subtract 15 years: remaining total=251y 11m 10d.

Sum=266 years 11 months 10 days; remaining mean≈13 years 3 months 4 days.

13. Calculate the four means from coded deviations.

(i) $D = X-140$: $\bar{x}$
$= 140+\frac{500}{10}$
$= 190.$
(ii) U=$\frac{x-130}{6}$: mean U=$\frac{-150}{15}$=−10, so $\bar{x}$=130+6(−10)=70.
(iii) $D = x-25$: $\bar{x}$
$= 25+\frac{300}{20}$
$= 40.$
(iv) $U = \frac{x-120}{5}$: $\bar{x}$
$= 120+5(\frac{60}{100})$
$= 123.$

14. Compare Haris (50,55,70,85,90), Maham (75,60,60,45,53), Minal (80,77,66,42,48).

Haris mean=$\frac{350}{5}$=70; Maham mean=$\frac{293}{5}$=58.6; Minal mean=$\frac{313}{5}$=62.6.

Haris receives Rs. 1,000 because 70 is the highest average.

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15. D=X−20 has D=−6,−4,−2,0,2,4,6 with frequencies 1,3,6,20,26,12,2. Find the mean.

$\Sigma f=70$
$\Sigma fD = -6-12-12+0+52+48+12$
$= 82.$
$\bar{x} = 20+\Sigma fD/\Sigma f$
$= 20+\frac{82}{70}$
$= 21.1714.$
Mean≈21.17.

16. For quiz points 45,51,58,61,74,48,46,50, use D=x−58.

$\Sigma x=433, n=8$ and $\Sigma D=433-8(58)=-31$. Mean$=58-\frac{31}{8}=54.125\approx 54.13$ points.

17. Find weighted mean cost/kg for rice 10 kg@96, flour 12@48, ghee 4@190, sugar 3@49, mutton 2@650.

Total quantity=31 kg. Total cost=960+576+760+147+1300=3,743.
Weighted mean$=\frac{3743}{31}$=Rs. 120.74 per kg.

18. Weighted mean for quantities 5,3,2,6 and costs 35,5,13,18 thousand rupees.

Weighted mean$=\frac{5\times35+3\times5+2\times13+6\times18}{5+3+2+6}=\frac{324}{16}$=Rs. 20.25 thousand.

19. Budgets for five years are 5,7,8,6,7 million rupees. Find the average.

Mean$=\frac{5+7+8+6+7}{5}=\frac{33}{5}$=Rs. 6.6 million.

20. Marks 78,65,80,90,85,72 have respective weights 5,4,2,3,2,4.

Weighted sum=390+260+160+270+170+288=1,538; total weight=20.
Weighted mean$=\frac{1538}{20}$=76.9 marks.

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